

Status: ✓ SAFE — governing safety factor 17.90 (Euler buckling; both modeled checks ≥ 2.0)

Design parameters

Bridge type	Warren truss, 6 bays
Span	36 m
Height	6 m
Load mode	Moving vehicle — current position
Vehicle load and position	120 kN at 1.0% of span (0.36 m from the left support)
Total applied load (W)	120.0 kN
Member cross-section	8500 mm ² (85 cm ²), solid square
Lateral bracing	Every 1 bay; longest top-chord unbraced interval 6.00 m
Material	Structural Steel (yield 350 MPa, E = 200 GPa)

Stress and buckling analysis

Analysis method	Method of joints (ideal pin-jointed truss)
Left support reaction	118.8 kN upward
Right support reaction	1.2 kN upward
Maximum bending moment	39.6 kN·m
Max absolute chord force	6.6 kN
Max absolute web-member force	14.8 kN
Max stress (force / area)	1.7 MPa
Yield safety factor	201.58
Buckling safety factor	17.90
Governing safety factor	17.90
Governing compression member	DL0; effective length 6.71 m; Euler capacity 264.1 kN (pinned ends, K = 1)

Quantities — one planar truss

Total member length	146.5 m
Estimated mass	9775 kg
Estimated material cost	\$58651

Current design reasoning

Prediction: I predict that a thicker section will improve the buckling margin because bending stiffness increases.

Observation: The whole crossing met the target. A diagonal changed from tension to compression as the vehicle moved.

Current design crossing test

Lowest safety factor 2.15 at 8.3% of span; buckling governs at DL0.

57 static vehicle positions tested with 120 kN. This records the crossing test separately from the current-position analysis above. Vehicle dynamics and deflection are not checked.

Saved design comparison

Saved inputs are recalculated with the current classroom model. Material cost and mass cover one planar truss. Each vehicle safety factor describes the saved position; completed crossing evidence is listed separately.

Crossing investigation

Saved geometry	warren truss; 36 m span × 6 m height; 6 bays
Saved material and bracing	Structural Steel; 8500 mm ² ; bracing every 1 bays
Saved applied load	120 kN vehicle at 1.0% of span; total 120.0 kN
Recalculated result	SAFE; governing SF 17.90; yield 201.58; buckling 17.90
Saved design quantities	\$58651; 9775 kg; Method of joints
Comparison with current design	Same span, layout, and applied load. Current material cost change: +\$. Compare safety margins before choosing a design.
Saved crossing evidence	Lowest safety factor 2.15 at 8.3% of span; buckling governs at DL0. 57 positions tested.

Prediction: I predict that a thicker section will improve the buckling margin because bending stiffness increases.

Observation: The whole crossing met the target. A diagonal changed from tension to compression as the vehicle moved.

Honest limit of this analysis: Member forces are solved for an ideal planar, pin-jointed truss. Yield and Euler buckling use solid square sections and pinned ends. Quantities cover one planar truss and exclude the deck, second truss, lateral-bracing members, connections, and waste. Material dead weight is not added to the applied load. Vehicle results describe only the current position unless an explicitly labeled crossing test is included. Deflection, fatigue, dynamics, and connection capacity are not checked. This classroom model is for learning, not construction.

Sources: NGSS Lead States (2013) · Hibbeler, R.C. (2017), Structural Analysis · AASHTO LRFD Bridge Design Specifications · Petroski, H. (1994), Design Paradigms (Tacoma Narrows + Tay Bridge + Hyatt). Printed from AlloFlow STEAM Lab.